

Medicinal Chemistry I
Final examination

Name Alfoalem Araba

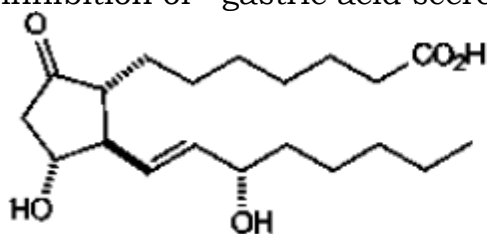
ID.No. HSR/4959/06

Time allowed 2 hrs

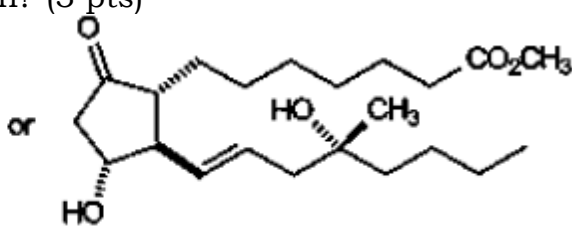
Maximum of 40 %

I. Answer the following questions briefly in the space provided

1. Explain briefly which compound has a longer duration of action for inhibition of gastric acid secretion? (3 pts)



A

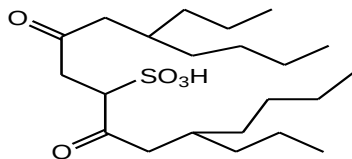


B

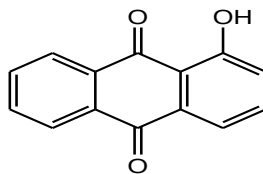
A Generic name Prostaglandin E1

B Generic name Misoprostol

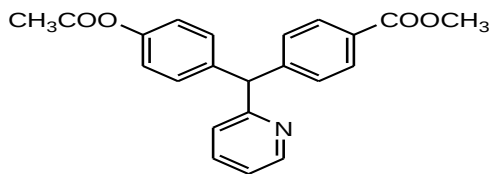
2. Match the following laxatives with their types (classes), PROVIDE the generic name when requested (3 pts)



Generic name _____



Generic name _____

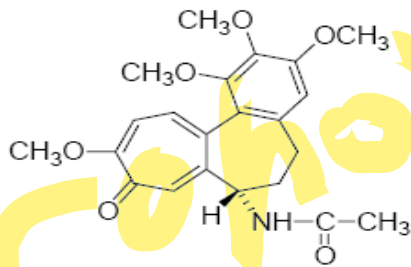


Generic name _____

- A. Irritant**
- B. Surfactant**
- C. Bulk laxative**

3. Write Briefly the bodies response for metabolic acidosis (2 pts)

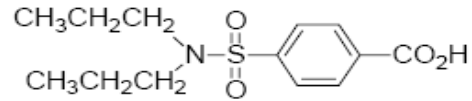
4. Which compound is more likely to be used to relieve the symptoms of gout by increasing the excretion of Uric acid?(2 pts)



Colchicine
Generic Name

colchicine

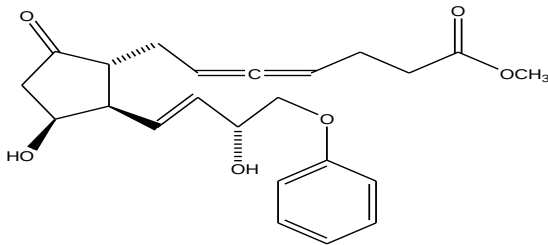
or



Generic Name

probenecid

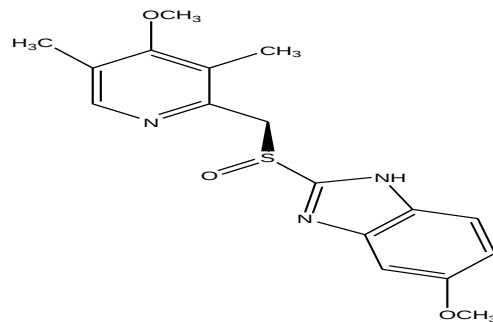
5. Indicate BRIEFLY and SPECIFICALLY the difference if any in the MECHANISM OF ACTION between the following two drugs .when requested provide the generic NAME . (3 pts)



Generic Name _____

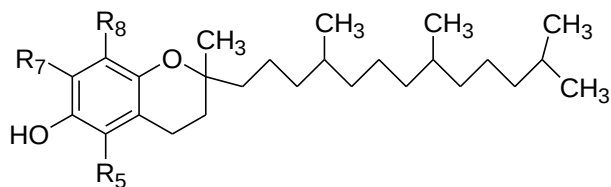
1

2 Omeprazole _____



Generic Name _____

6. Based on the following general structure of tocopherols, fill the vacant spaces of the table (3 marks).

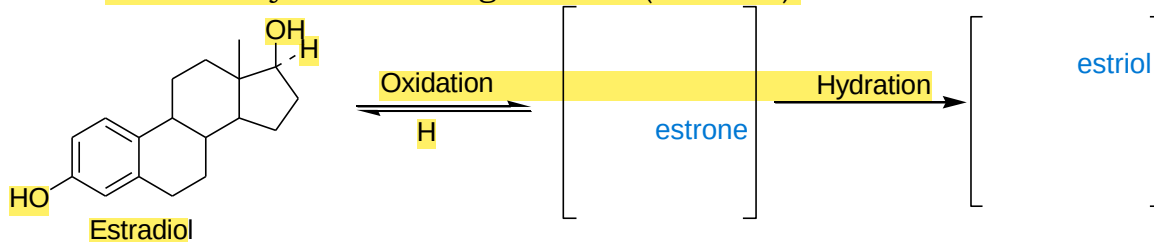


Name of Compound	R5	R7	R8
γ- tocopherol	hydroxyl	Methyl	Methyl
β- tocopherol	Methyl	Hydroxyl	Methyl
α- tocopherol	Methyl	Methyl	Methyl
δ- tocopherol	Hydroxyl	Hydroxyl	Methyl

Arrange the compounds in question 9 **in the order of increasing** biological activity (1 mark).

_____ α- < _____

7. Give the structures of two natural estrogens produced from estradiol by the following reaction (2 marks).



Choose the best Answer (2 pts)

1. The drug illustrated below:

	I Is metabolized to two molecules of 5-aminosalicylic acid II Is used to treat ulcerative colitis III Is metabolized by bacteria in the large intestine to a cyclooxygenase inhibitor
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- a) I only
- b) III only
- c) I and II only
- d) **II and III only**
- e) I, II, and III

Answer _____

2. One of the following is an intracellular ion?

- a. Chlorine
- b. Phosphate
- c. Bicarbonate
- d. Sodium

3. One of the following is false about electrolytes?

- a. Phosphate is important for calcium metabolism
- b. Potassium and magnesium have effect on nerve transmission
- c. Hypokalemia can occur during diarrhea
- d. Thiazide diuretics can be used for the treatment of hypokalemia
- e. None

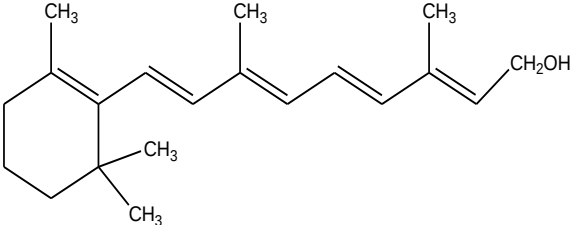
4. One of the following is **not correct** about **fat soluble vitamins**

- a. Soluble In lipid
- b. Are usually associated with lipids of foods and are absorbed from the intestine with this dietary lipids
- c. They are stored in the liver and thus converted by the liver.
- d. **Are soluble in water**
- e. None

5. One of the following is **different** from the others

- a. Vitamin A
- b. Vitamin C**
- c. Vitamin E
- d. Vitamin K

6. The drug illustrated below is

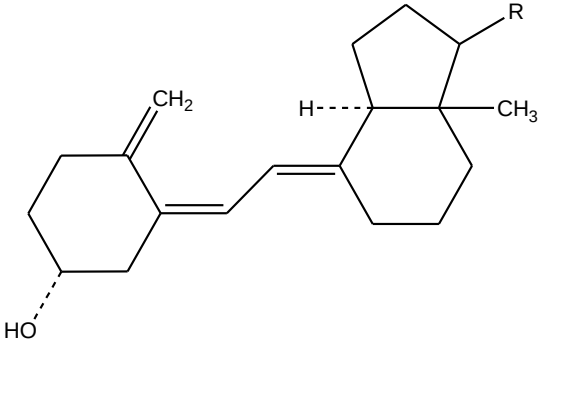
	a. All trans retinol b. growth vitamin c. obtained from β-carotenes d. 11-mono cis or Δ^4 cis retinol
---	--

- A. I only
- B. II & III only
- C. I & III only
- D. II ,III &IV Only
- E. I,II & III only**

7. One of the following is **not correct** about **the SAR of vitamin A**

- a. The conjugated double bond systems found in vit.A and β -carotene are necessary for activity
- b. partial or complete reduction of the double bonds produce active compounds**
- c. The β -ionone ring of retinal or the dehydro β -ionone ring is essential for activity
- d. None

8. The drug illustrated below is

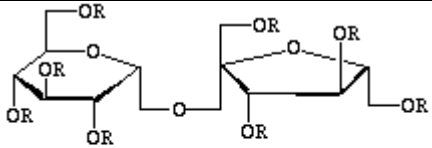
	a. Sunshine vitamins b. Obtained by irradiation of yeast ergosterol c. possess significant antiarachitic properties
---	---

- a. I only
- b. I & II only**

c. II & III only

d. I, II & III

9. The drug illustrated below is :

 sucralfate (Carafate®) $R = \text{SO}_3\text{Al}(\text{OH})_3 \cdot (\text{H}_2\text{O})_2$	a. Protects the gastric mucosa by forming a sticky viscous gel b. adheres to mucosal tissue c. has inherent acid-neutralizing activity
--	---

a. I Only

b. II Only

c. I & II only

d. None

10. All are steroidal Estrogens but one:

a. Estradiol

b. Diethylstilbestrol

c. Mestranol

d. Ethinyl estradiol

e. **None**

11. In natural Estrogens;

a. Ring A is benzene

b. All are Steroidal

c. Can be used as contraceptive

d. Contain cyclopentanoperhydrophenanthrene nucleus

e. All

12. The following progestins are derived either from testosterone or nortestosterone except:

a. Norethindrone

b. Provera

c. Norgestrel

d. Dimethisterone

e. Norethynodrel

13. One of the following does not contain cyclic (ring) in its structure

a. Vitamin B5

b. Vitamin B6

c. Vitamin B9

d. Vitamin B2

e. Vitamin B3

14. A vitamin known for its antioxidant activity is:
- a. Vitamin E
 - b. Vitamin C
 - c. Ascorbic Acid
 - d. All**
15. A statement which is false about the biosynthesis of Steroids is:
- a. Cholesterol is a common precursor
 - b. Hydrocortisone can be directly produced from progesterone
 - c. Testosterone and 17β - estradiol can be synthesized from progesterone
 - d. Aldosterone is a precursor steroid for progesterone
 - e. Aromatization is ring A of testosterone produces 17β - estradiol

Bonus!!!

1. Give some examples of enzymatic and non enzymatic antioxidant!!(3 marks)

Enzymatic

Glutathione peroxidase(GSH)

Catalase

Superoxide Dismutase(SOD)

Non enzymatic

Vitamin E

Vitamin c

Vitamin a